Topology, Domain Shift and Efficiency in Retinal Vessel Segmentation: A Controlled Source-Only Study of TG-LiteNet

CSD361 – Introduction to Machine Learning, Final Project Report

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***Abstract*—A vessel segmenter for rural retinal screening must keep thin vessels *connected*, work on cameras it was never trained on and run on modest hardware. We study this three-way trade-off in one controlled, source-only protocol (training on DRIVE only; STARE, CHASE_DB1 and HRF held out) with *TG-LiteNet*, a compact depthwise-separable encoder–decoder with a soft-clDice topology loss, an auxiliary skeleton head, domain-generalisation (DG) augmentation with a consistency loss, and Monte-Carlo-dropout uncertainty. A staged ablation (5 seeds per stage), a $2\times2\times2$ factorial design, variants, plain compact U-Nets and evaluation-only controls are scored with Dice, clDice, Betti error, size, latency and uncertainty–error correlation. TG-LiteNet has 0.21 M parameters and 5.5 GFLOPs (U-Net: 31.04 M) and reaches an unseen Dice of 0.756 and clDice of 0.833, versus 0.762 and 0.835 for the U-Net. At the fixed threshold, topology supervision more than halves the raw unseen Betti-0 error of the U-Net (75.0 to 31.7) and adds 0.007 clDice at a cost of 0.009 Dice, but removing small components brings every model to about 5, and with a threshold tuned on DRIVE validation the topology-trained models have the highest Dice (stage C 0.782, plain U-Net 0.757). DG augmentation adds about 0.01 unseen Dice. The depthwise backbone saves parameters, not time: plain compact U-Nets match its Dice at equal or lower CPU latency, and a base-16 U-Net is also better connected, so we would deploy that. Neither the skeleton head, the consistency loss nor dropout adds anything measurable. MC-dropout ranks pixel errors moderately ($\rho_{px} \approx 0.31$–0.34), but image-level rejection is not reliably better than random. No target image or label is used for fitting, tuning or checkpoint selection; the targets were inspected while designing the analyses, so they are held-out, not untouched.**

***Keywords*—retinal vessel segmentation, topology-aware loss, domain generalization, lightweight networks, uncertainty estimation, DRIVE**

I. Introduction

Segmenting the retinal vascular tree from fundus photographs supports the measurement of vessel calibre, tortuosity and branching used in screening for diabetic retinopathy, glaucoma and hypertensive retinopathy [1]. Since U-Net [2] such networks are trained and tested on small public datasets, typically DRIVE [3], STARE [4], CHASE_DB1 [5] and HRF [6]. Overlap scores hide topological errors: a severed centreline barely changes Dice but breaks the connectivity that biomarkers rely on [7]–[9].

Motivation. A model for a low-resource screening programme must satisfy three constraints *simultaneously*: preserve vessel *topology*; tolerate camera and population *shift*, ideally without target labels [12], [13]; and be small and fast enough for *edge* hardware [14]. For a metric m the *target gap* is $\Delta_m = m(\text{source}) - m(\text{target})$.

Objectives. (O1) A reproducible *source-only* pipeline for topology-aware vessel segmentation; (O2) the contribution of each component (topology loss, DG augmentation, compact backbone, uncertainty) to accuracy, topology, cross-domain gap and cost in one controlled ablation; (O3) whether MC-dropout uncertainty can triage unreliable predictions, especially on unseen domains.

Approach and contributions. We study *TG-LiteNet*, a depthwise-separable encoder–decoder with a mask head and an auxiliary skeleton head. Contributions: (1) a staged ablation and a full $2\times2\times2$ factorial study (backbone $\times$ topology loss $\times$ DG augmentation) with repeated seeds and confidence intervals; (2) a unified evaluation of Dice/IoU, clDice, Betti-0/1 error, the source-to-target gap, parameters, FLOPs, CPU/GPU latency and uncertainty quality *together*; (3) an uncertainty-based triage analysis at pixel and image level for every domain; (4) controls that test whether the gains survive simple alternatives (small-component removal, validation-tuned thresholds, plain compact U-Nets, removing single ingredients); and (5) open code that regenerates every number. We claim neither a new loss or architecture family nor state-of-the-art accuracy.

II. Related Work

Base paper: clDice. Shit et al. [7] compare the skeleton of a prediction with the ground-truth mask (topology precision) and the skeleton of the ground truth with the predicted mask (topology sensitivity); their harmonic mean is clDice. They prove that maximising clDice preserves topology up to homotopy equivalence for binary tubular structures and give

TABLE I

EVIDENCE REPORTED BY THE CLOSEST REVIEWED WORKS (✓: REPORTED, ∼: PARTIAL, –: NOT REPORTED; N/V: NOT VERIFIABLE). COLUMNS: TOPOLOGY LOSS, TOPOLOGY METRICS, TOTAL DEPLOYMENT COST, UNCERTAINTY. PROTOCOL: SO = SOURCE-ONLY, MS = MULTI-SOURCE DG, SD = SINGLE-DATASET EVALUATION, TTA, FT = FINE-TUNING ON LABELLED TARGET DATA.

Work	Prot.	Loss	Topo.	Cost	Unc.
clDice, SRL [7], [9]	none	✓	✓	∼	–
DGSSA, WaveRNet [15], [16]	MS	–	–	–	–
Lightweight [12], [14], [19]	SO	∼	–	✓	–
TFFM [20]	SD	✓	✓	–	–
TopoTTA [21]	TTA	∼	✓	n/v	–
TopoLoRA-SAM, TAPDecoderFT [11], [22]	FT	✓	✓	∼	–
Topology-aware unc. [23]	–	–	∼	n/v	✓
TG-LiteNet (this work)	SO	✓	✓	✓	✓

a differentiable *soft-clDice* based on iterated soft morphological opening. cbDice [8] observes that clDice favours large vessels ("diameter imbalance") and adds radius and boundary information; Skeleton Recall Loss [9] computes the skeleton once, offline, with cheap CPU operations; earlier work used persistent homology [10]. None is evaluated for source-only robustness under shift.

Domain generalisation and adaptation. "Cross-dataset" covers source-only transfer, multi-source DG, test-time adaptation (TTA) and fine-tuning with target labels. DGSSA [15] synthesises vessel trees and styles (multi-source); WaveRNet [16], Hu et al. [17] and GraphSeg [18] use frequency, Hessian or graph priors. A benchmark [13] finds that, with enough data, a plain U-Net generalises well.

Lightweight networks. Galdran et al. [12] show that U-Nets with ∼34K and W-Nets with ∼68K parameters are competitive; DSAE-Net [19] (78K parameters, skeleton-distance loss) and LightVesselNet [14] (75K parameters) report cross-dataset Dice but no connectivity metrics or uncertainty.

Topology under shift; uncertainty. TopoTTA [21], TopoLoRA-SAM [22] and TAPDecoderFT [11] report clDice/Betti after test-time or fine-tuned transfer, with large or unspecified models and no latency; TFFM [20] adds topology-aware feature fusion but is evaluated on a single dataset. Topology-aware uncertainty [23] scores whole structures; MC dropout [24] is a cheap pixel-level alternative whose quality degrades under dataset shift [25].

Research gap. Table I summarises what the closest works report. Among the works we reviewed, none evaluates a compact *source-only* model for topology, total cost and uncertainty quality in one controlled protocol with repeated runs; we fill that gap with a trade-off *study*, not a new method.

III. PROBLEM FORMULATION AND RESEARCH QUESTIONS

Given a fundus image I with field-of-view (FOV) mask F, predict a binary vessel mask $M \in \{0,1\}^{H\times W}$. Models are trained on DRIVE only; STARE, CHASE_DB1 and HRF are never used for fitting, loss-weight or threshold tuning, or checkpoint selection (they were inspected while designing the analyses, Sec. VII-A). We ask:

- **RQ1 (topology).** Does topology supervision (soft-clDice + skeleton head) improve clDice and Betti error relative to Dice+BCE without an unacceptable loss of overlap?
- **RQ2 (generalisation).** Does lightweight style-plus-structure augmentation with a consistency loss reduce Δ_m from DRIVE to the unseen targets?
- **RQ3 (trade-off).** What accuracy and topology cost does a $\leq$2M-parameter budget incur relative to the savings in parameters, FLOPs and latency?
- **RQ4 (uncertainty).** Does MC-dropout uncertainty rank high-error cases and support selective rejection, more so on unseen domains than on the source domain?

IV. PROPOSED METHODOLOGY

A. Overview

Fig. 1 shows the pipeline. A fundus image is rescaled so that its FOV has a fixed width, reduced to the CLAHE-enhanced, standardised green channel, and passed through an encoder–decoder with a mask head (and, during training only, a skeleton head). The loss combines region, pixel, topology and consistency terms. At test time, T stochastic forward passes give a mean probability map and a variance map used as uncertainty.

B. Preprocessing and scale normalisation

Cameras differ strongly in resolution (HRF images are $\sim$6$\times$ wider than DRIVE's), so the pixel width of vessels, to which a convolutional model is very sensitive, changes with the dataset. Every image is rescaled so that its FOV is W_F=527 px wide (DRIVE's median), which uses neither labels nor any statistic of the target *dataset*. We then take the green channel (best vessel contrast), replace the outside of the FOV by the FOV mean, apply CLAHE [26] with a fixed tile size in pixels, standardise inside the FOV, and set the outside to zero. The same deterministic function is used in training and testing.

C. Architectures

Baseline U-Net [2]: four 2$\times$2 max-pool stages, double 3$\times$3 conv+BN+ReLU blocks, transposed-convolution up-sampling, base width 64 ($\approx$31.04 M parameters).

TG-LiteNet backbone. Every 3$\times$3 convolution is replaced by a depthwise 3$\times$3 followed by a pointwise 1$\times$1 convolution (BN, ReLU6), as in MobileNets [27]. A block has two such layers and a 1$\times$1 residual shortcut. Encoder widths are $(w, 2w, 4w, 8w, 8w)$ with w=16; the decoder uses bilinear up-sampling, skip concatenation and one separable block per level. A 1$\times$1 mask head outputs vessel logits; an auxiliary 1$\times$1 skeleton head (training only, no inference cost) predicts the 1-px-dilated skeleton of the label. Spatial dropout (p=0.1) after the bottleneck and each decoder block is used only in stage E. The width w is the single capacity knob; Section VII sweeps it.

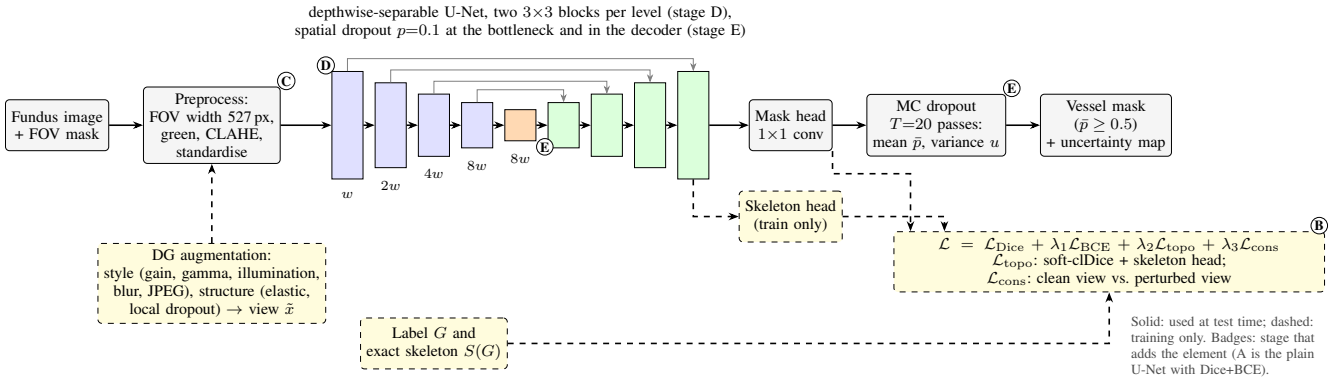


Fig. 1. TG-LiteNet. A fundus image is normalised to a fixed FOV width and passed through a depthwise-separable U-Net with a mask head; T stochastic passes give the segmentation and an uncertainty map. Skeleton head, DG augmentation and the loss (dashed) exist only during training; one fixed model is evaluated on every target.

D. Objective

For a prediction $P = \sigma(z) \in [0,1]^{H \times W}$, label G and FOV F, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{topo}} + \lambda_3 \mathcal{L}_{\text{cons}}, \qquad (1)$$

with all terms computed inside F. $\mathcal{L}_{\text{Dice}}$ and $\mathcal{L}_{\text{BCE}}$ are the usual soft-Dice and binary cross-entropy.

Topology term. With $S(\cdot)$ the (soft) morphological skeleton, topology precision and sensitivity are $T_{\text{prec}} = \frac{\sum S(P) \odot G}{\sum S(P)}$, $T_{\text{sens}} = \frac{\sum S(G) \odot P}{\sum S(G)}$, and

$$\mathcal{L}_{\text{clDice}} = 1 - \frac{2\, T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}. \qquad (2)$$

$S(P)$ is the differentiable skeleton obtained by iterating soft erosion/opening (10 iterations, enough for vessels up to 10 px radius at our scale); $S(G)$ is the exact skeleton of the (augmented) label, computed in the data loader. The skeleton head adds $\mathcal{L}_{\text{skel}} = \text{Dice} + \text{BCE}$ against the dilated skeleton, so $\mathcal{L}_{\text{topo}} = \mathcal{L}_{\text{clDice}} + \gamma \mathcal{L}_{\text{skel}}$. As supplementary variants we re-implemented Skeleton Recall [9], $\mathcal{L}_{\text{SR}} = 1 - \sum P \odot \mathcal{T}(S(G)) / \sum \mathcal{T}(S(G))$ with $\mathcal{T}$ a 3×3 dilation, and the radius- and boundary-aware cbDice [8] (cl-MIN-Dice variant; the label's skeleton and distance map are exact, the prediction's are the soft skeleton and a soft-erosion distance map, so values may differ numerically from the authors' code). cbDice is evaluated as a metric for every model and tested as a loss variant (Sec. VII).

Consistency term (DG). For each patch we build a perturbed view $\tilde{x}$ (below) that is pixel-aligned with the clean view x: $\mathcal{L}_{\text{cons}} = \frac{1}{|F|} \sum_F \left(\sigma(f(\tilde{x})) - \text{sg}[\sigma(f(x))]\right)^2$, where sg stops gradients. Both views are also supervised.

E. Domain-generalisation augmentation

All stages use the same geometric augmentation (random crop at a FOV pixel, flips, free rotation, scaling in $[0.8, 1.25]$). Stage C onwards additionally applies, *only at training time*: **style** — gain/gamma/contrast jitter, a smooth multiplicative illumination field, blur, noise and JPEG compression of the green channel before CLAHE; and **structure** — an elastic deformation applied identically to both views and the label, and local intensity dropout (view 2 only). Because CLAHE and standardisation follow the jitter, gain, gamma and contrast are largely neutralised by design; the effective style perturbations are illumination, blur, noise and JPEG. A DG stage processes two views per patch, i.e. twice the forward passes of stages A/B, and is deliberately much simpler than synthesising vessel trees [15].

F. Uncertainty and triage

With dropout kept active (BatchNorm in evaluation mode), T passes give $p_t = \sigma(f_t(x))$, the mean $\bar{p} = \frac{1}{T} \sum_t p_t$ and the variance $u = \frac{1}{T} \sum_t p_t^2 - \bar{p}^2$ [24]. We compare u with the predictive entropy $H(\bar{p})$, which is also available for deterministic models. For triage, an image is scored by its mean u inside F; the images (or pixels) with the largest score are rejected first.

V. Experimental Setup

A. Datasets, preprocessing and exploratory analysis

Table II lists the datasets. DRIVE provides 20 training and 20 test images; we split the training images once (seed 2026) into 16 for training and 4 for validation (model selection and, if used, hyper-parameter selection), and use the 20 test images as the in-domain test set. STARE (20), CHASE_DB1 (28) and HRF (45) are used only for testing. Ground truth is the first human observer. HRF and (when available) official DRIVE masks provide the FOV; for DRIVE (mirror), STARE and CHASE_DB1 the FOV is estimated by thresholding, keeping the largest component and eroding it by 1% (checked visually, Fig. 2). The vessel fraction is 9–13% in every dataset, i.e. about 1:8 class imbalance; there are no missing values or categorical attributes in this task. The exploratory statistics of Table II quantify the domain shift: the mean green-channel intensity ranges from ≈60 (CHASE_DB1, HRF) to ≈110 (STARE), and while the vessel calibre after FOV normalisation is ≈3.5 px for DRIVE, STARE and CHASE_DB1, it is only ≈1.4 px for HRF: HRF's reference annotations are thin relative

TABLE II

DATASETS (ALL STATISTICS MEASURED ON THE FILES USED IN OUR EXPERIMENTS). FOV = FIELD-OF-VIEW WIDTH, USED TO NORMALISE THE SCALE OF EVERY IMAGE; CALIBRE = MEAN VESSEL WIDTH (2× DISTANCE TRANSFORM ON THE LABEL SKELETON) AT NATIVE RESOLUTION AND AFTER NORMALISATION TO A 527 PX FOV; G = GREEN-CHANNEL MEAN±STD INSIDE THE FOV (0–255). SOURCE = TRAIN (16) / VALIDATION (4) / TEST (20 OFFICIAL TEST IMAGES); TARGETS ARE TEST-ONLY.

Dataset	#	Size (px)	Vessel %	FOV (px)	Calibre (px)	G	Role
DRIVE [3]	40	565×584	12.9	527	3.5→3.5	97±17	source
STARE [4]	20	700×605	10.5	638	4.3→3.5	109±24	target
CHASE_DB1 [5]	28	999×960	10.3	899	6.1→3.6	62±26	target
HRF [6]	45	3504×2336	9.1	3262	8.8→1.4	60±17	target

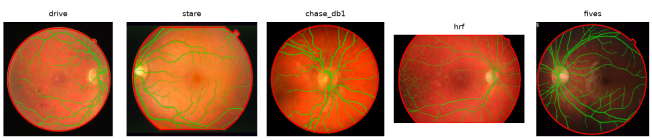


Fig. 2. Exploratory view of the datasets: vessel labels (green) and FOV outlines (red). Camera, colour balance, field of view and resolution differ markedly between datasets, which is the domain shift studied here.

to its field of view, so HRF measures an annotation-style shift in addition to the camera shift (a no-rescale evaluation is in Fig. 4). Sources and checksums are recorded in `data/MANIFEST.json`.

B. Implementation and procedure

Everything is implemented in Python with PyTorch [28] (2.11.0+cu128), OpenCV, scikit-image [29] (Zhang–Suen thinning [30] for skeletons, connected components, Euler number) and scikit-learn. Training ran on Kaggle (Tesla T4, PyTorch 2.11.0+cu128; 37 runs) and Colab (NVIDIA L4, cu130; 33 runs: seeds 3–4 of A–E, seed 2 of the 3-seed configurations, all compact-U-Net and control runs) at three code commits that differ only in analysis code and added experiments (`run_manifest.csv`); runs are statistically, not bitwise, reproducible. All stages share one optimiser, schedule and data pipeline (AdamW, learning rate 10^{-3}, weight decay 10^{-4}, 200 warm-up iterations then cosine decay, 5000 iterations of 8 patches of 256^2 px, λ_1=1, λ_2=λ_3=0.5, 10 soft-skeleton iterations, CLAHE clip 2 with 64-px tiles; `configs/base.yaml`) and differ only in one switch each: A plain U-Net (Dice+BCE), B + topology loss, C + DG augmentation with consistency, D + compact backbone, E + dropout and MC averaging (Fig. 1). We train each stage A–E with 5 seeds and each additional configuration (factorial cells, loss variants, widths, controls) with 3 seeds, and report mean±std; differences between configurations are paired over shared seeds (Student-t on the per-seed differences, 95% half-widths; `stage_deltas.csv`, `factorial_deltas.csv`).

C. Evaluation metrics

All metrics are computed inside the FOV on the binarised prediction (threshold 0.5, fixed before seeing any target) against the *native-resolution* ground truth: the model runs at the normalised scale and its probability map is resampled (bilinear) to the label resolution, because down-scaling thin labels fragments them and inflates Betti-0. **Accuracy:** Dice, IoU, sensitivity, specificity, accuracy and AUC (histogram-based above 2 MP). **Topology:** clDice (exact skeletons), cbDice and the Betti-0/Betti-1 errors, i.e. the absolute difference in the number of connected components (β_0, 8-connectivity) and loops ($\beta_1 = \beta_0 - \chi$). **Generalisation:** $\Delta_m = m(\text{DRIVE}) - m(\text{target})$; "unseen" is the macro average over the three targets. **Cost:** parameters, FLOPs (2×MACs) and median latency for one 512×512 image (CPU with 1 and 4 threads, GPU). **Uncertainty:** Spearman correlation between pixel uncertainty and misclassification (ρ_{px}) and between image-mean uncertainty and 1−Dice (ρ_{img}), risk–coverage curves and the area under the image-level curve (AURC) with random and oracle references.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Main results (staged ablation)

Table III gives the staged ablation and Fig. 3 the per-dataset values. Differences below are means over seeds with paired 95% half-widths in brackets (training randomness; the same seed identifier forms a pair). We also compare models on the same test images (`paired_image_deltas.csv`): the paired difference of each image's seed-averaged score, macro-averaged over STARE, CHASE_DB1 and HRF like the tables, with a stratified bootstrap interval (test-set sampling only). With 5 seeds (3 for supplementary runs) and no multiplicity correction both kinds of interval are descriptive.

Topology (A→B, RQ1). The topology loss lowers the raw unseen Betti-0 error from 75.0 to 31.7 ($-43.3\,[\pm16.0]$, on every target) and raises unseen clDice by 0.007 (±0.004), at a cost of 0.009 unseen Dice (±0.004) and 0.003 on DRIVE (±0.002). On the 93 unseen images Dice is lower for 92% and clDice higher for 96%. Unseen AUC moves by only −0.001 (±0.001), so ranking quality is nearly unchanged. It does not reduce loop errors: unseen Betti-1 moves only from 22.5 to 21.0 and is worse for the compact models (24.4). RQ1 therefore holds for centreline overlap and the raw fragment count, not for area overlap or loops; the checks below show that the fragment count is mostly removable by simple cleanup and that much of the Dice cost is threshold calibration.

Generalisation (B→C, RQ2). DG augmentation raises unseen Dice by 0.008 (±0.005; higher for 81% of the images) and shrinks the DRIVE-to-unseen gap from 0.074 to 0.065 (-0.009 ± 0.004). CHASE_DB1 sensitivity rises from 0.879 to 0.927, while unseen clDice (+0.004, ±0.004) and Betti-0 (−3.8, ±3.4) change little. RQ2 holds for overlap, only marginally for topology.

Compact backbone (C→D, RQ3). Shrinking from 31 M to 0.21 M parameters costs 0.004 Dice on DRIVE (±0.002), 0.005 unseen Dice (±0.003; lower for 72% of the images) and clearly more topology: unseen clDice −0.014 (±0.004) and raw Betti-0 +16.8 (±10.3), mostly on HRF, whose thin annotations reward capacity (sensitivity 0.891 to 0.862). The gap is unchanged (0.065). Stage E (dropout, MC averaging) differs from D by at most 0.002 Dice (lower for 81% of the

TABLE III

STAGED ABLATION (MEAN±STD OVER SEEDS). SOURCE = DRIVE TEST; UNSEEN = MACRO AVERAGE OVER STARE, CHASE_DB1 AND HRF; ΔDICE = DICE(DRIVE)−DICE(UNSEEN). COST: ONE 512×512 IMAGE, BATCH 1, CPU = 4 THREADS, KAGGLE MACHINE; E (MC) NEEDS T PASSES. BEST PER COLUMN IN BOLD.

	Source (DRIVE)		Unseen domains							Latency (ms)	
Stage	Dice	clDice	Dice	clDice	cbDice	β_0 err	ΔDice	Params (M)	GFLOPs	CPU	GPU
A	**0.830**±0.001	0.830±0.003	**0.762**±0.006	0.835±0.002	**0.732**±0.005	74.99±14.36	0.068±0.006	31.04	384.7	2915	91.1
B	0.827±0.001	0.845±0.001	0.753±0.004	0.843±0.003	0.728±0.003	31.69±2.27	0.074±0.004	31.04	384.7	2886	89.6
C	0.826±0.001	**0.849**±0.001	0.761±0.001	**0.847**±0.001	0.729±0.002	**27.93**±2.11	**0.065**±0.001	31.04	384.7	2770	89.3
D	0.822±0.001	0.838±0.003	0.757±0.003	0.832±0.002	0.722±0.004	44.71±6.49	0.065±0.003	0.21	5.5	405	12.2
E (det.)	0.822±0.001	0.839±0.003	0.755±0.005	0.833±0.002	0.720±0.005	41.95±2.84	0.066±0.005	0.21	5.5	429	12.2
E (MC)	0.822±0.001	0.838±0.003	0.756±0.004	0.833±0.002	0.721±0.005	42.66±2.98	0.066±0.004	0.21	5.5 × 20	429 × 20	12.2 × 20

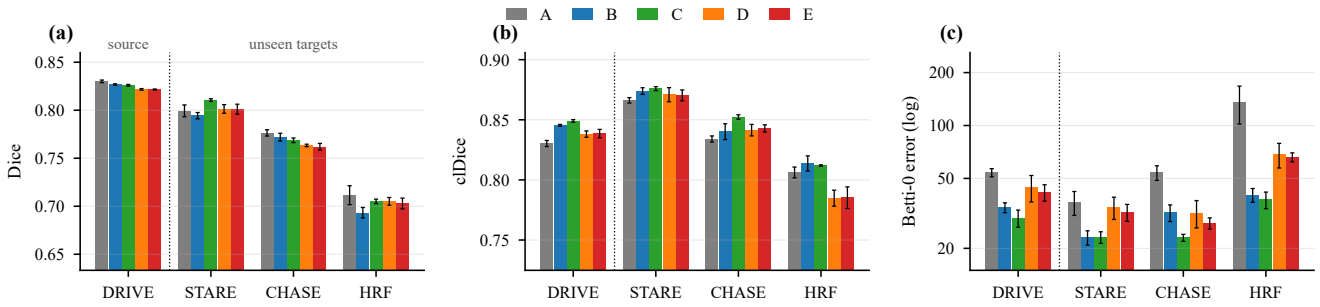


Fig. 3. Per-dataset results of stages A–E: Dice, clDice and Betti-0 error (log scale; lower is better). Bars are means over seeds, whiskers the standard deviation over seeds; DRIVE is the source domain.

images) and not in clDice or Betti-0: dropout brings no benefit, only the uncertainty map. Overall (A→E), unseen Dice is 0.007 lower (±0.007), unseen clDice 0.003 lower (±0.003) and raw unseen Betti-0 32 lower (±20).

Classification view. Table IV gives the pixel confusion matrix. On unseen data all stages are recall-heavy (recall 0.90–0.92) but over-segment: precision is only 0.66–0.67 against 0.81 for TG-LiteNet on DRIVE, and false positives (4.5–4.8% of the FOV) outnumber false negatives (0.8–1.0%) about five to one. The dominant error is spurious thin vessels and fragments, not missed vessels, which fits the Betti-0 counts.

Fragments, threshold and scale. Betti-0 error is dominated by tiny floating fragments: the ground truth has 3–12 components per image, the predictions 25–187. Fig. 4 shows evaluation-only sensitivity checks with the cut-off k and a threshold chosen per model on the four DRIVE-validation images (`postproc_choices.csv`; 5 seeds for A–E, 3 for the others). Removing components smaller than k px^2 at the 527-px FOV scale (k=100–200; the FOV width is estimated from its area) brings the unseen Betti-0 error of every model to 4.6–5.0, including the plain U-Net (75.0 to 4.7), with Dice up by at most 0.002 and clDice down by at most 0.004: the raw-count advantage of the topology loss is not specific to it. After cleanup the clDice gain of the topology loss alone is +0.006 (±0.008, not significant) and that of topology plus DG +0.012 (±0.005). The tuned thresholds are higher for the topology-trained models (0.50–0.65, plain U-Net 0.45–0.50), which over-segment (recall 0.90, precision 0.66 at 0.5): tuning raises their unseen Dice by 0.015–0.020 (B 0.771, C 0.782, D 0.772, E 0.771) but not the plain U-Net's (−0.006, ±0.007). At tuned thresholds the topology loss therefore no longer costs Dice but gains it (A→B $+0.014 \pm 0.008$, A→C $+0.025 \pm 0.006$), so the Dice cost at 0.5 is mostly calibration; the DG-only U-Net gains little (0.777, $+0.006 \pm 0.013$) and C is $+0.002 \pm 0.002$ above it (p=0.05, 3 seeds) with +0.007 clDice and 46 fewer raw Betti-0. The thresholds come from four images and are post-hoc, so all main results keep 0.5. Without FOV rescaling (seeds 3–4) the plain U-Net collapses (unseen Dice 0.768 to 0.637, HRF 0.50) while D and E hold (0.757 to 0.750, 0.756 to 0.751), but only A, D and E were run at native scale, so we cannot say what gives the lite models this robustness.

TABLE IV

PIXEL-LEVEL CONFUSION MATRIX (VESSEL = POSITIVE, THRESHOLD 0.5): TP, FP, FN, TN IN % OF THE FOV PIXELS, MEANS OVER IMAGES AND SEEDS. TOP: TG-LITENET (E) PER DATASET. BOTTOM: U-NET (A) AND TG-LITENET ON THE UNSEEN AVERAGE (ALL STAGES: `CONFUSION.CSV`). F1 EQUALS THE PER-IMAGE DICE.

	TP	FP	FN	TN	Precision	Recall	F1	Accuracy
DRIVE	10.91	2.59	2.14	84.37	0.809	0.839	0.822	0.953
STARE	9.65	3.91	0.82	85.62	0.712	0.921	0.801	0.953
CHASE_DB1	9.39	4.93	0.88	84.80	0.655	0.915	0.762	0.942
HRF	7.88	5.27	1.25	85.60	0.597	0.864	0.703	0.935
A, unseen avg.	8.94	4.45	1.02	85.59	0.669	0.897	0.762	0.945
E (det.), unseen avg.	8.97	4.70	0.98	85.34	0.655	0.900	0.755	0.943

B. *Efficiency and the trade-off (RQ3)*

TG-LiteNet (Fig. 5) has 147× fewer parameters and 70× fewer FLOPs than the U-Net, but is only 6.8× faster on four

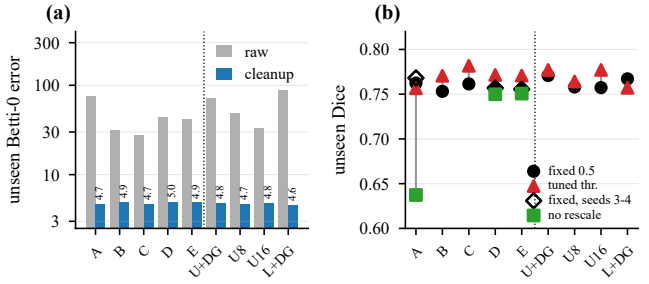


Fig. 4. Evaluation-only checks (unseen average; A–E: 5 seeds; U+DG, U8, U16, L+DG: 3 seeds). (a) Betti-0 error before and after removing components below a size chosen on DRIVE validation (cleaned value printed). (b) Dice at the fixed threshold 0.5, at the per-model threshold chosen on DRIVE validation and without FOV rescaling (A, D, E; seeds 3–4, hollow marker: fixed Dice of the same seeds).

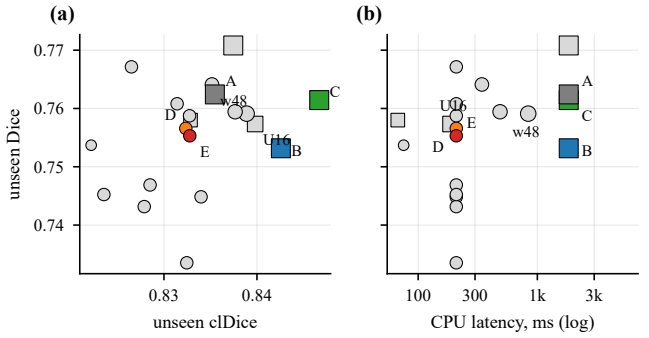


Fig. 5. Unseen Dice against (a) unseen clDice and (b) CPU latency (laptop, log scale). Marker area grows with the parameter count; squares: U-Net, circles: lite; letters: stages A–E; U16: plain U-Net with base width 16; w48: lite width 48; grey: other configurations (Table VI).

CPU threads in the Kaggle runs (429 vs 2915 ms), 7.5× on the T4 (12.2 vs 91.1 ms) and 8.8× in a controlled laptop benchmark (209 vs 1834 ms): depthwise convolutions have low arithmetic intensity, so FLOPs overstate the speed-up. Against plain compact U-Nets the picture reverses (Table VI): a base-8 U-Net with the same FLOPs runs in 67 ms, 3.1× faster than the lite model, with the same Dice, and a base-16 U-Net (1.9 M parameters, within the 2 M budget) is as fast (187 ms) with better clDice and Betti-0. The depthwise-separable backbone therefore saves parameters, not time, and on this evidence a base-16 U-Net is the better choice for CPU-only screening hardware. Against the U-Net, TG-LiteNet gives up 0.008 DRIVE Dice, 0.006 unseen Dice and 0.003 unseen clDice; this loss is not larger on unseen domains (identical gap 0.065 for C and D). MC dropout multiplies the cost by T=20.

C. Uncertainty (RQ4)

Pixel-level uncertainty (per-domain values in `results/tables/uncertainty.csv`) is moderately informative: ρ_{px} is 0.31 on DRIVE and slightly higher on the unseen domains (0.32 STARE, 0.34 CHASE_DB1, 0.34 HRF), so the expected increase under shift (RQ4) is present but small. Entropy is marginally better than MC variance at

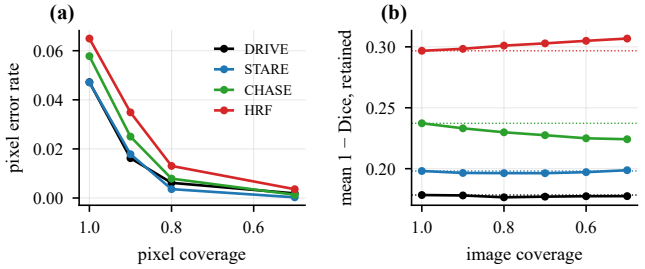


Fig. 6. Risk–coverage of the MC-dropout variance of TG-LiteNet: (a) pixel error rate after rejecting the most uncertain pixels, (b) mean 1−Dice of the retained images when the most uncertain images are rejected (dotted: random rejection).

TABLE V
FULL FACTORIAL DESIGN (BACKBONE × TOPOLOGY × DG). UNSEEN = MACRO AVERAGE OVER THE THREE TARGETS. STAGES A–D (U-NET PLAIN, U-NET+TOPO, U-NET+TOPO+DG, LITE+TOPO+DG) HAVE 5 SEEDS, THE OTHER FOUR CELLS 3.

Backbone	Topo	DG	Src Dice	Unseen Dice	Unseen clDice	ΔDice
U-Net	–	–	0.830±0.001	0.762±0.006	0.835±0.002	0.068±0.006
U-Net	–	✓	0.831±0.000	0.771±0.001	0.837±0.000	0.060±0.001
U-Net	✓	–	0.827±0.001	0.753±0.004	0.843±0.003	0.074±0.004
U-Net	✓	✓	0.826±0.001	0.761±0.001	0.847±0.001	0.065±0.001
Lite	–	–	0.823±0.001	0.745±0.011	0.824±0.001	0.078±0.011
Lite	–	✓	0.824±0.000	0.767±0.002	0.827±0.001	0.057±0.002
Lite	✓	–	0.821±0.002	0.747±0.007	0.829±0.004	0.074±0.009
Lite	✓	✓	0.822±0.001	0.757±0.003	0.832±0.002	0.065±0.003

pixel level, variance at image level (ρ_{img} 0.151 vs 0.129 on DRIVE). Rejecting the 10% most uncertain pixels (Fig. 6a) lowers the pixel error from 4.7% to 1.6% on DRIVE and from 6.5% to 3.5% on HRF. At image level (Fig. 6b) the evidence is weak: AURC equals random on DRIVE and STARE, is better by 4% on CHASE_DB1 (ρ_{img}=0.48) and *worse* than random on HRF (ρ_{img}=−0.17). With 20–45 images per domain these statistics are imprecise, but they do not support referring whole images with MC variance [25]; with the T=20 cost they do not justify MC dropout here, and deep ensembles or temperature scaling (not run) would be the next options. The curves rank errors but say nothing about calibration (no ECE or Brier score).

D. Qualitative results and error analysis

In Fig. 7 (one image per dataset, seed 0, windows chosen where A and E differ, so the differences are exaggerated) the errors of all three models are mostly thin red (false positive) or blue (missed) fragments along the finest vessels, and spurious red segments such as the ones on CHASE_DB1 appear in all three; the large vessels are essentially error-free. On HRF the compact models miss more capillary-scale vessels than the U-Net (mean HRF sensitivity 0.864 for E vs 0.889 for A), which matches their lower HRF clDice. The uncertainty maps of E mark vessel borders and thin segments in general rather than the red and blue errors in particular, consistent with the weak image-level triage above.

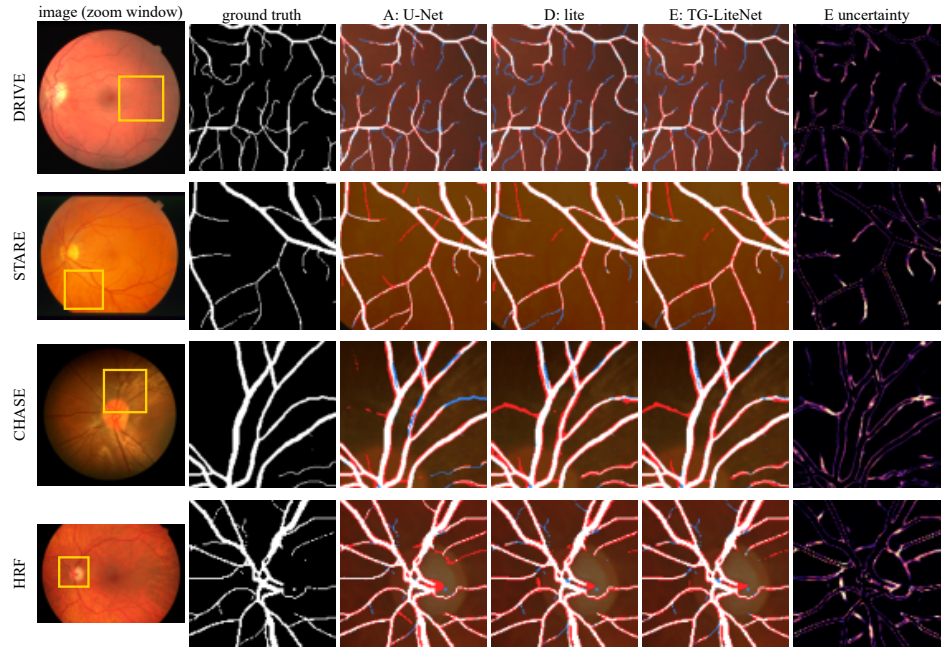


Fig. 7. Qualitative results (seed 0, one image per dataset). The yellow window, where stages A and E disagree most, is shown enlarged: ground truth, error maps of A, D and E (white: true positive, red: false positive, blue: false negative; threshold 0.5) and the MC-dropout uncertainty of E (scaled per image).

VII. Ablation Analysis

Table V separates the factors (unseen averages; effects paired over shared seeds). *Topology loss* lowers the raw Betti-0 error in all four pairs (U-Net 75→32 and 73→28, lite 86→46 and 88→45; every paired interval excludes zero), raises clDice by 0.005–0.009 and changes unseen Dice by $+0.002$ to -0.011, so it still helps a $\leq$2 M-parameter network in raw Betti-0 and clDice, with the cleanup caveat. *DG augmentation* adds 0.008 to 0.022 unseen Dice, narrows the gap (0.068→0.060 for the plain U-Net) and leaves Betti-0 unchanged. *Backbone*: the compact network loses 0.004–0.013 Dice and 0.011–0.014 clDice and adds 15–20 raw Betti-0 error in every cell. At the fixed threshold the highest unseen Dice come from DG without topology loss (U-Net 0.771; lite 0.767, $+0.009 \pm 0.006$ over the baseline but with worse clDice, -0.011 ± 0.006); the full TG-LiteNet recipe is a connectivity-oriented operating point. In Table VI (top) the default soft-clDice has the best Dice: cbDice and clDice+Skeleton Recall lower the raw Betti-0 error slightly (44.7 to 39.1 and 36.5, within the seed spread) at 0.012 ($\pm$0.013) and 0.023 ($\pm$0.011) unseen Dice, and Skeleton Recall alone is worse in clDice (-0.006, $\pm$0.002). Neither the skeleton head nor the consistency loss helps Dice or clDice (without head: 0.761 and 0.831; without consistency: 0.759 and 0.833; stage D: 0.757 and 0.832); raw Betti-0 is higher by 10 ($\pm$3) and 8 ($\pm$9) without them, the only trace of a benefit. The plain compact U-Nets with the stage-D recipe match the lite model in Dice (base 8, 0.49 M parameters: 0.758; base 16, 1.9 M: 0.757) and the larger one is more connected (clDice 0.840 vs 0.832, paired $+0.006 \pm 0.003$; Betti-0 33.7 vs 44.7). In the width sweep, unseen Dice saturates near w=24 while clDice (0.822 to 0.839) and Betti-0 (61 to 30) keep improving up to w=48.

TABLE VI

Variants of stage D (unseen averages): topology loss (top), removed ingredients and plain compact U-Nets with the same recipe (middle), lite width (bottom). Stage D has 5 seeds, other rows 3. CPU = 4 threads, controlled laptop benchmark.

	Dice	clDice	β_0 err	Params (M)	CPU (ms)
soft-clDice + head (D)	0.757±0.003	0.832±0.002	44.7	0.21	209
cbDice	0.745±0.002	0.834±0.002	39.1	0.21	209
Skeleton Recall	0.743±0.006	0.828±0.001	55.3	0.21	209
clDice + Skel. Recall	0.734±0.001	0.832±0.003	36.5	0.21	209
D without head	0.761±0.006	0.831±0.001	53.8	0.21	209
D without consistency	0.759±0.005	0.833±0.002	52.2	0.21	209
plain U-Net, base 8	0.758±0.006	0.833±0.001	48.9	0.49	67
plain U-Net, base 16	0.757±0.005	0.840±0.001	33.7	1.94	187
width w=8	0.754±0.005	0.822±0.001	61.1	0.06	76
width w=16 (D)	0.757±0.003	0.832±0.002	44.7	0.21	209
width w=24	0.764±0.002	0.835±0.003	40.0	0.46	343
width w=32	0.759±0.005	0.838±0.003	34.3	0.81	487
width w=48	0.759±0.005	0.839±0.002	30.1	1.81	838

A. Limitations and threats to validity

(i) Only 16 training images; seed-to-seed variation can match small gains (stage A's unseen Dice ranges from 0.755 to 0.769 over five seeds). Stages A–E use 5 seeds, other configurations 3, the no-rescale check 2. (ii) The unseen average weights three very different datasets equally; raw Betti counts are dominated by HRF. (iii) The no-rescale evaluation covers only A, D and E. (iv) Labels are the first observer only; four of five FOV sets are estimated. (v) Soft-clDice, Skeleton Recall and cbDice [8] are our re-implementations. (vi) Latency is eager-PyTorch, not an edge measurement, and

varied up to 2× on shared cloud CPUs, hence the laptop benchmark. (vii) The targets were never used for fitting, loss-weight or threshold tuning or checkpoint selection, but we saw their results while designing the ablations, so they are held-out rather than untouched; comparisons are uncorrected, and the cleanup size and per-model thresholds (four validation images) are post-hoc. (viii) The DG stage bundles augmentation, a consistency term and a second view; the controls remove only the consistency term and the skeleton head. (ix) Uncertainty is ranked, not calibrated. (x) Source-only evaluation is harder than multi-source or test-time adaptation, so absolute numbers are not comparable with such papers [15], [16], [21]. (xi) Checkpoints are not in the repository: the committed per-image results are reproducible by retraining, not by re-evaluation.

VIII. Conclusion and Future Work

We built and evaluated TG-LiteNet in a controlled source-only protocol (one fixed 0.5 threshold unless stated). (O1) The pipeline is source-only and every number comes from 70 trained models. (O2) The topology loss lowers the raw unseen Betti-0 error of the U-Net from 75 to 32 and raises clDice by 0.007 at 0.009 Dice, but simple cleanup brings every model to Betti-0 ≈5, so its lasting benefit is the clDice gain (RQ1); with a threshold tuned on DRIVE validation (5 seeds, post-hoc) the topology-trained U-Nets have the highest Dice (stage C 0.782, plain U-Net 0.757), so the Dice cost at 0.5 is mostly calibration. DG augmentation adds 0.008 unseen Dice and narrows the shift gap from 0.074 to 0.065 (RQ2); at the fixed threshold the best Dice is DG without topology loss (U-Net 0.771), but at tuned thresholds stage C is on par or slightly better (0.782 against 0.777) and better connected. The compact backbone costs 0.005 Dice, 0.014 clDice and 17 raw Betti-0 (RQ3). TG-LiteNet is 0.006 Dice below the baseline U-Net with 147× fewer parameters and 70× fewer FLOPs, but plain compact U-Nets with the same recipe match its accuracy (base 8) or also its connectivity (base 16: clDice 0.840) at equal or lower CPU latency, so the depthwise backbone saves parameters, not time, and a base-16 U-Net is the better deployment choice on this evidence. Removing the skeleton head or the consistency loss lowers neither Dice nor clDice and dropout adds nothing measurable. (O3) MC-dropout ranks pixel errors moderately (ρ_{px} 0.31–0.34), but image-level rejection is within 4% of random and worse on HRF, so at 20× the cost we would drop it. **Limitations** are listed in Sec. VII-A. **Future work:** a warm-up of the topology weight, deep ensembles or temperature scaling with calibration metrics, the official cbDice code, held-out FIVES, and INT8 measurements on a real edge device.

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